



USER MANUAL

Gocator 1300 Series: Non-Standard Model Specifications

Document revision: A

Copyright

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Information contained within this manual is subject to change.

This product is designated for use solely as a component and as such it does not comply with the standards relating to laser products specified in U.S. FDA CFR Title 21 Part 1040.

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Introduction

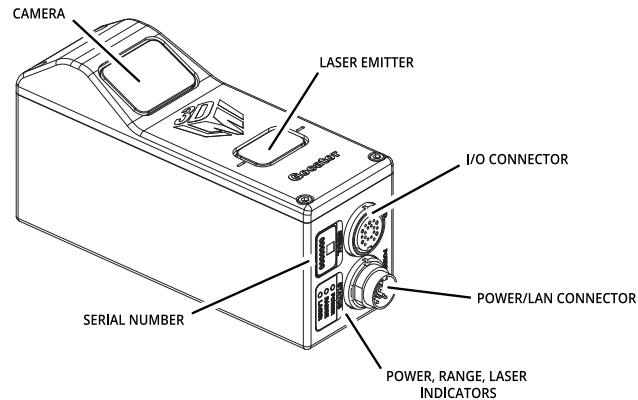
This document provides the specifications of the non-standard Gocator 1300 series models, including field of view / measurement range, dimension, and envelope drawings.

For information on safety and maintenance, setup, the Gocator web interface, protocols, job files, other hardware specifications, and more, refer to the Gocator 1300 Series user manual.

Hardware Overview

For information on Masters and cordsets, refer to the Gocator 1300 series user manual.

Top Mount Package



Item	Description
Camera	Observes laser light reflected from target surfaces.
Laser Emitter	Emits structured light for laser ranging.
I/O Connector	Accepts input and output signals.
Power / LAN Connector	Accepts power and laser safety signals and connects to 1000 Mbit/s Ethernet network.
Power Indicator	Illuminates when power is applied (blue).
Range Indicator	Illuminates when camera detects laser light and is within the target range (green).
Laser Indicator	Illuminates when laser safety input is active (amber).
Serial Number	Unique sensor serial number.

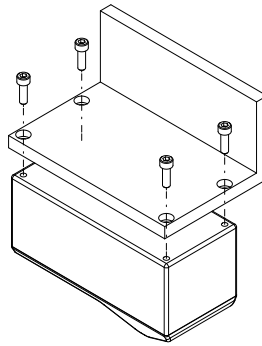
Installation

The following section provides mounting information.

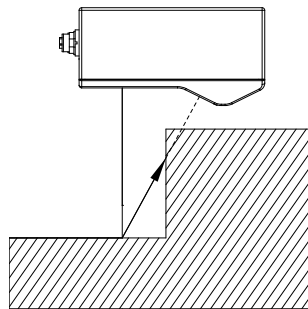
For information on grounding and orientation, refer to the Gocator 1300 user manual.

Mounting - Top Mount Package

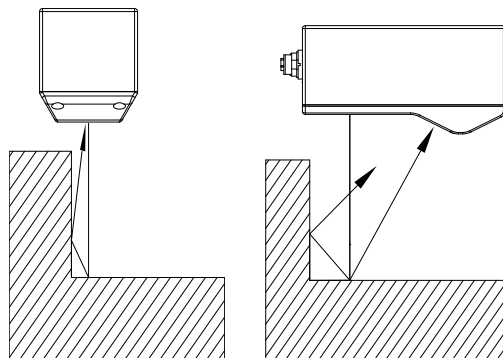
Sensors should be mounted using four M5 x 0.8 pitch screws of suitable length. The recommended thread engagement into the housing is 8-10 mm. Proper care should be taken in order to ensure that the internal threads are not damaged from cross-threading or improper insertion of screws.



Sensors should not be installed near objects that might occlude a camera's view of the laser.



Sensors should not be installed near surfaces that might create unanticipated laser reflections.





The sensor must be heat sunk through the frame it is mounted to. When a sensor is properly heat sunk, the difference between ambient temperature and the temperature reported in the sensor's health channel is less than 15° C.



Gocator sensors are high-accuracy devices. The temperature of all of its components must be in equilibrium. When the sensor is powered up, a warm-up time of at least one hour is required to reach a consistent spread of temperature within the sensor.

Specifications

The following sections provide the specifications of the non-standard Gocator 1300 models.

Gocator 1300 Series (Non-Standard Models)

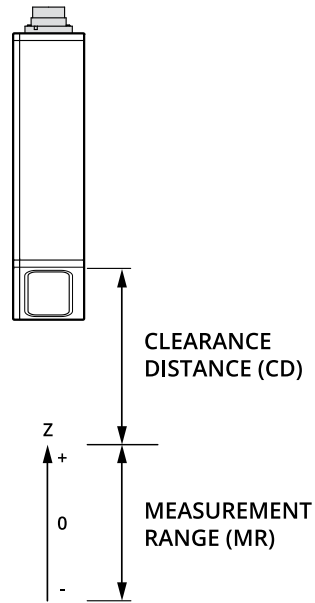
MODEL	1320	1340	1365	1370	1390
Clearance Distance	40	162.5	562	237.5	500
Measurement Range (MR) (mm)	20	95	375	412.5	2000
Linearity Z (+/- % of MR)	0.05	0.03	0.11	0.07	0.10
Linearity Z (+/- mm)	0.010	0.03	0.4	0.3	2.0
Resolution Z (mm)	0.0004 - 0.0004	0.0005 - 0.0010	0.0025 - 0.0040	0.0025 - 0.0070	0.0250 - 0.0600
Spot Size (mm)	0.11	0.37	1.80	0.90	2.60
Laser Classes	2M, 3B	2M, 3R	2M, 3R	2M, 3R	3R
Package Dimensions (mm)	Top Mount 49x75x162	Top Mount 49x75x162	Top Mount 49x75x232	Top Mount 49x75x162	Top Mount 49x75x289
Weight (kg)	0.8	0.8	1.1	0.8	1.4

ALL 1300 SERIES MODELS

Scan Rate	32kHz
Interface	Gigabit Ethernet
Inputs	Differential Encoder, Laser Safety Enable, Trigger
Outputs	2x Digital Output, RS-485 Serial, Selcom Serial, 1x Analog Output (4 - 20 mA)
Input Voltage (Power)	+24 to +48 VDC (13 Watts); Ripple +/- 10%
Housing	Gasketed Aluminum Enclosure, IP67
Operating Temp.	0 to 50° C
Storage Temp.	-30 to 70 ° C

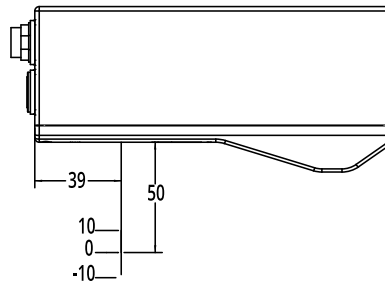
Specifications are preliminary; final specifications may vary. Resolution Z based on 128 samples averaging.

Specifications stated reflect standard laser classes; values will vary for other laser classes.

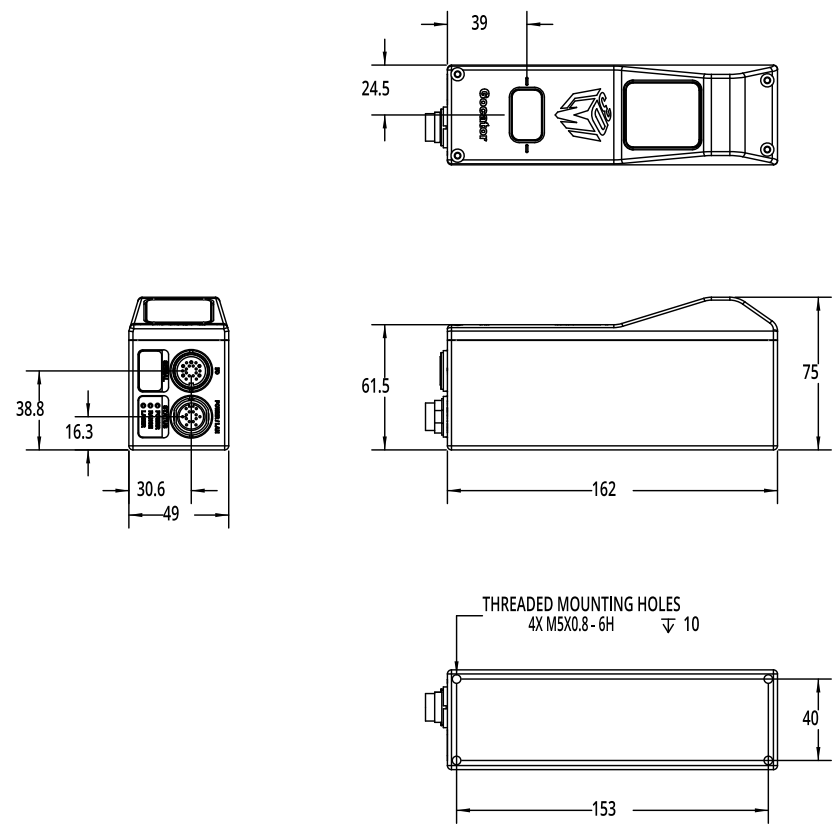


Gocator 1320 (Top Mount Package)

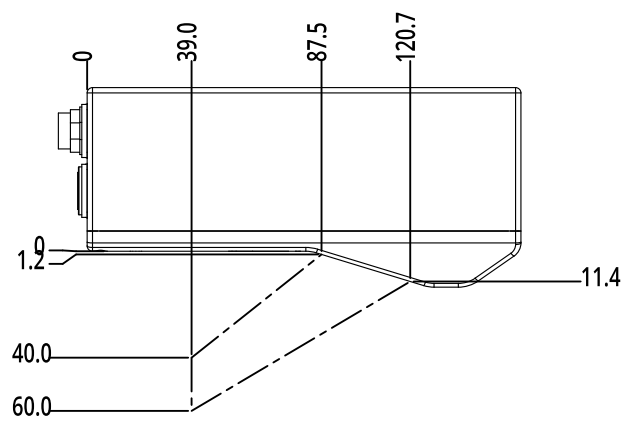
Field of View / Measurement Range



Dimensions

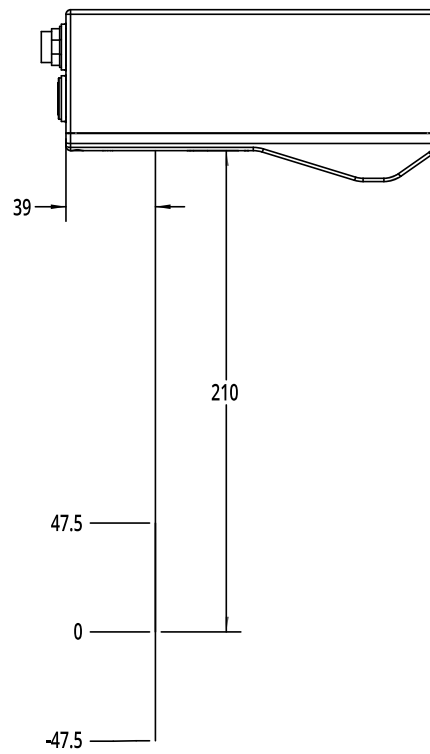


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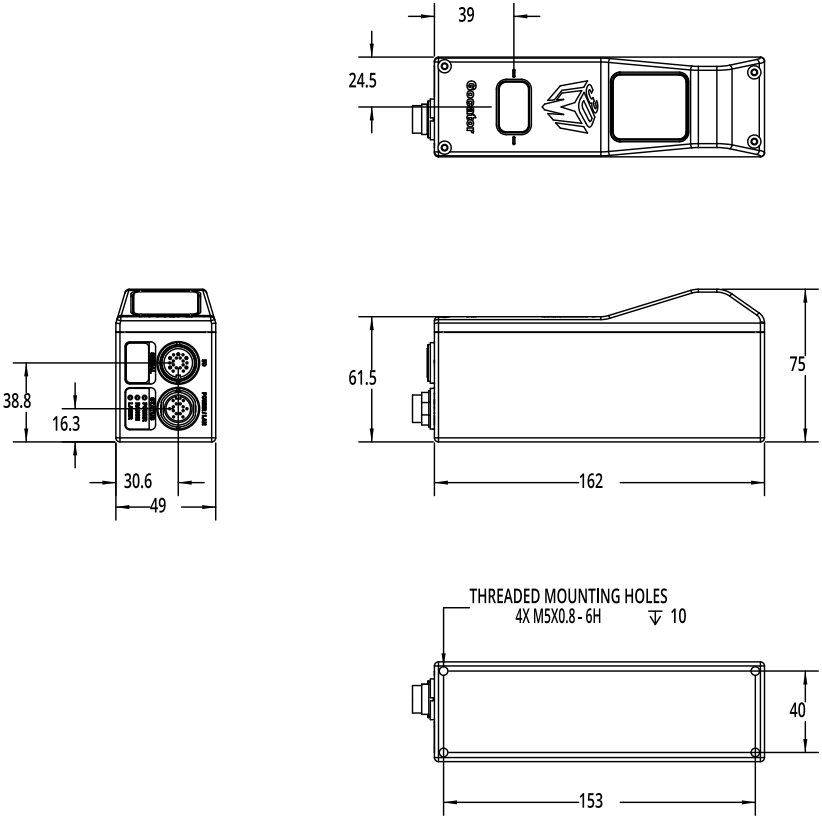


Gocator 1340 (Top Mount Package)

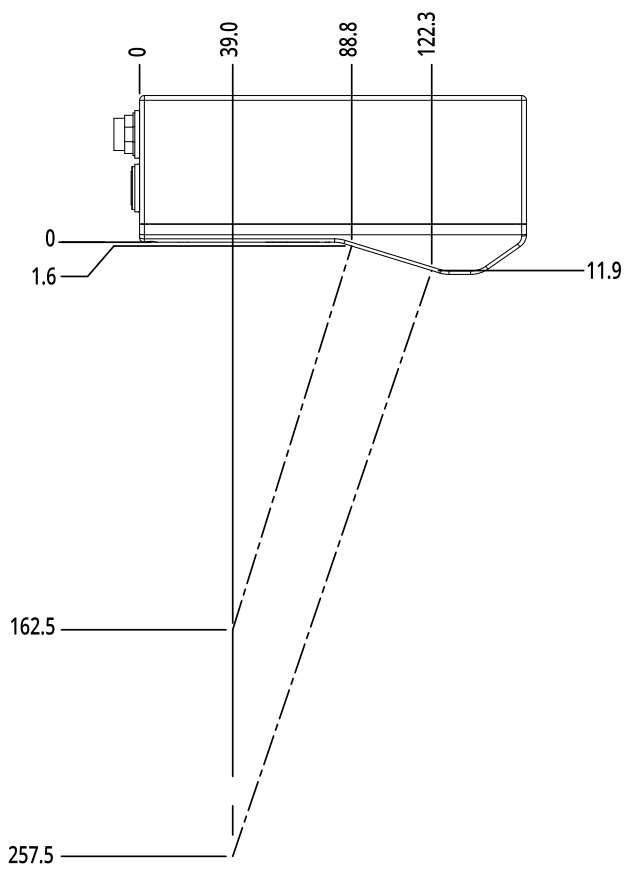
Field of View / Measurement Range



Dimensions

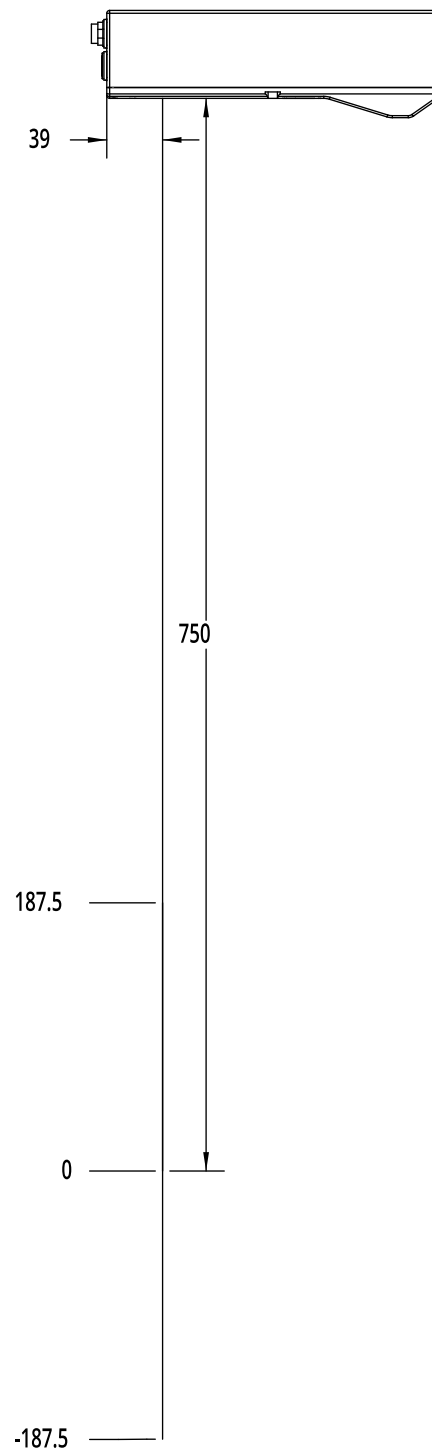


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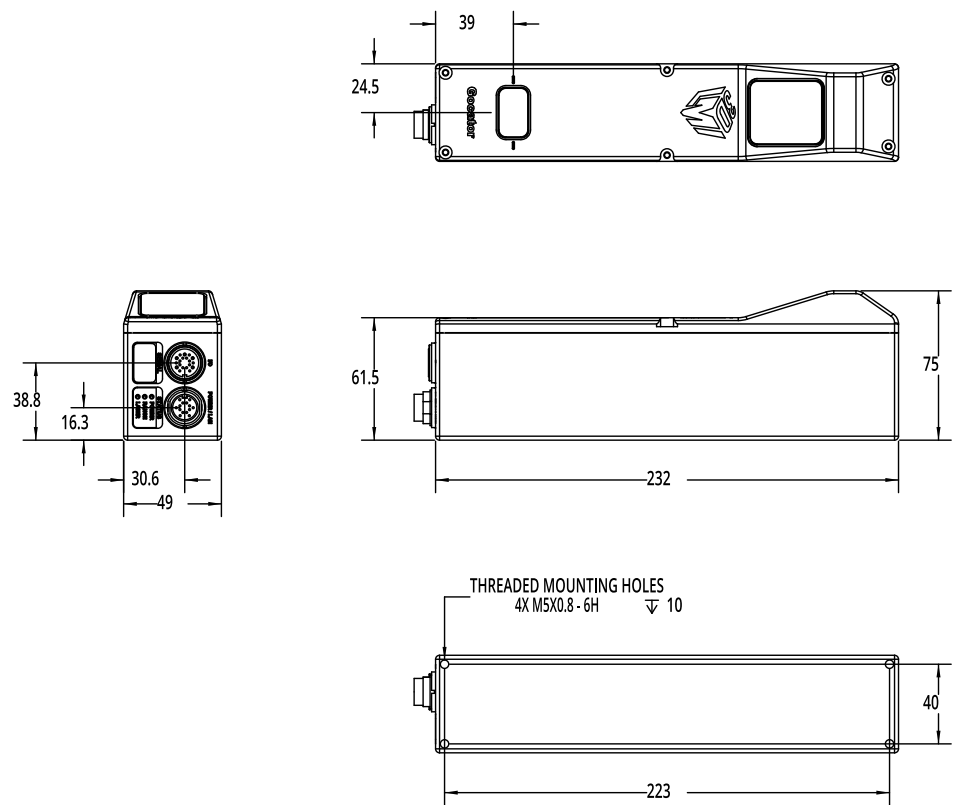


Gocator 1365 (Top Mount Package)

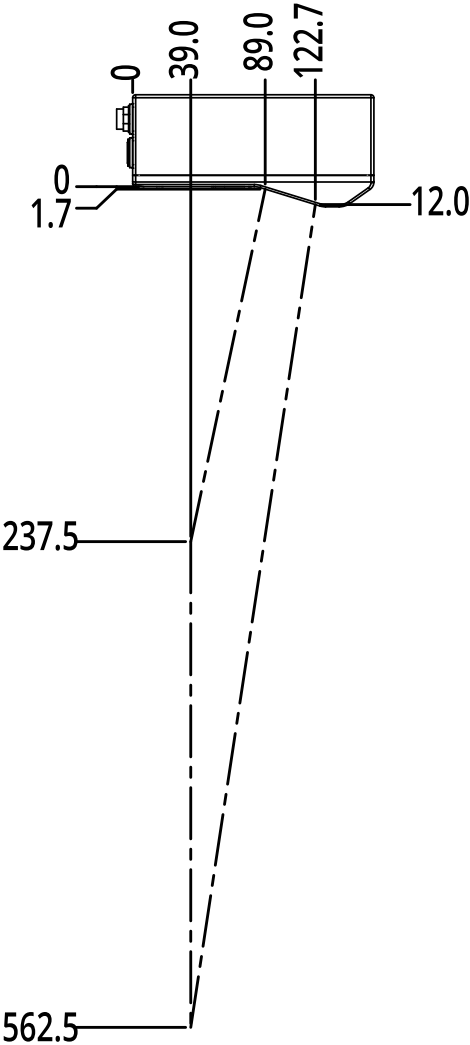
Field of View / Measurement Range



Dimensions

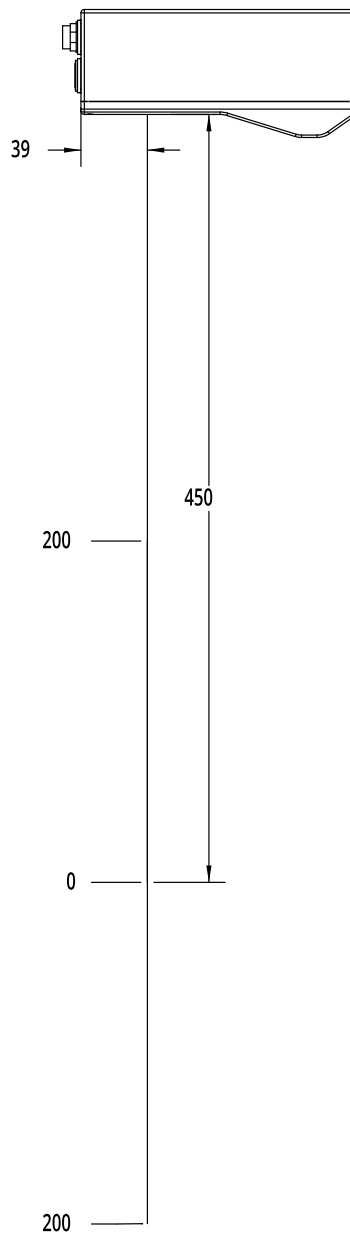


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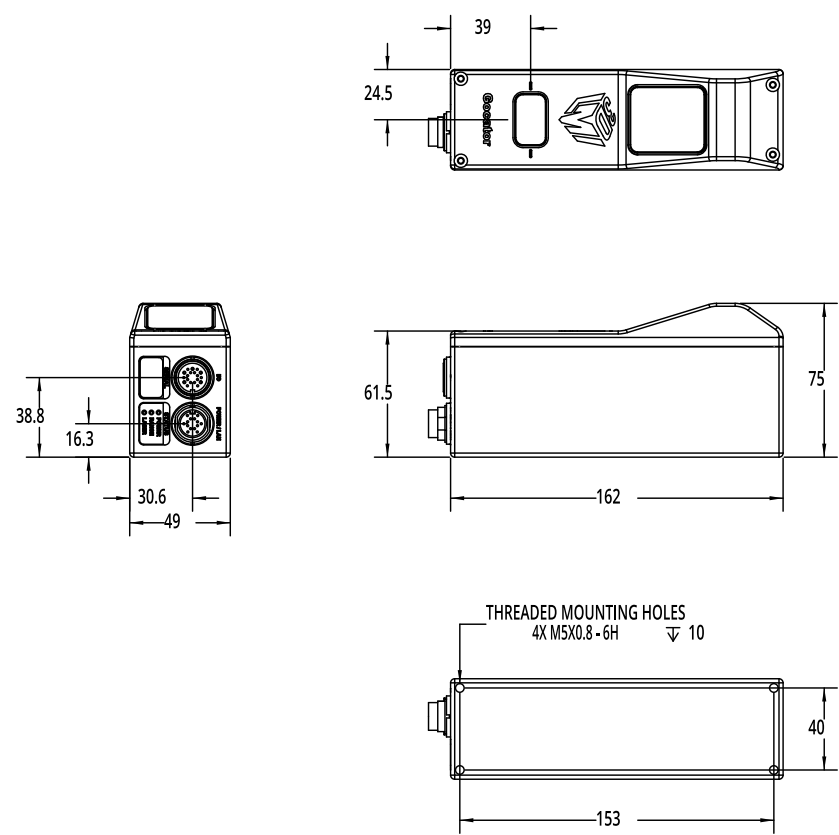


Gocator 1370 (Top Mount Package)

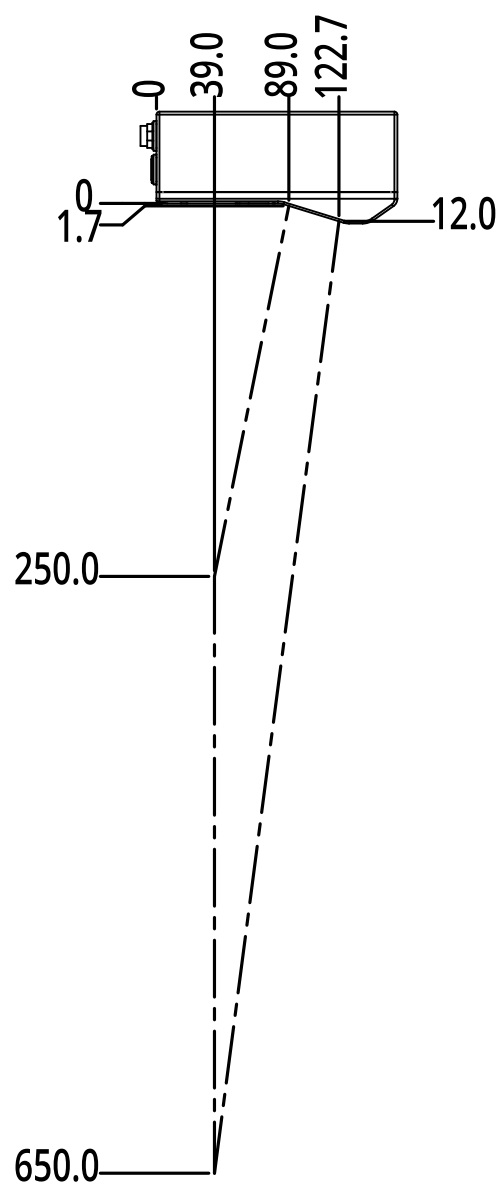
Field of View / Measurement Range



Dimensions

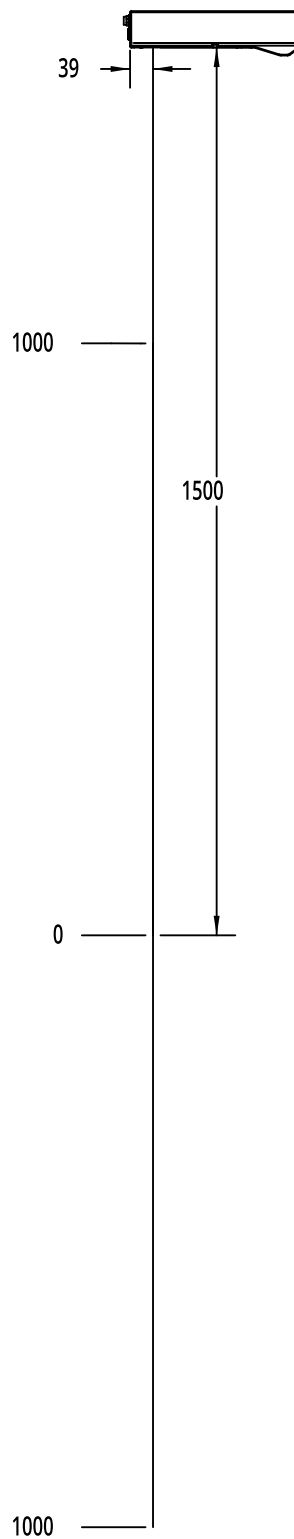


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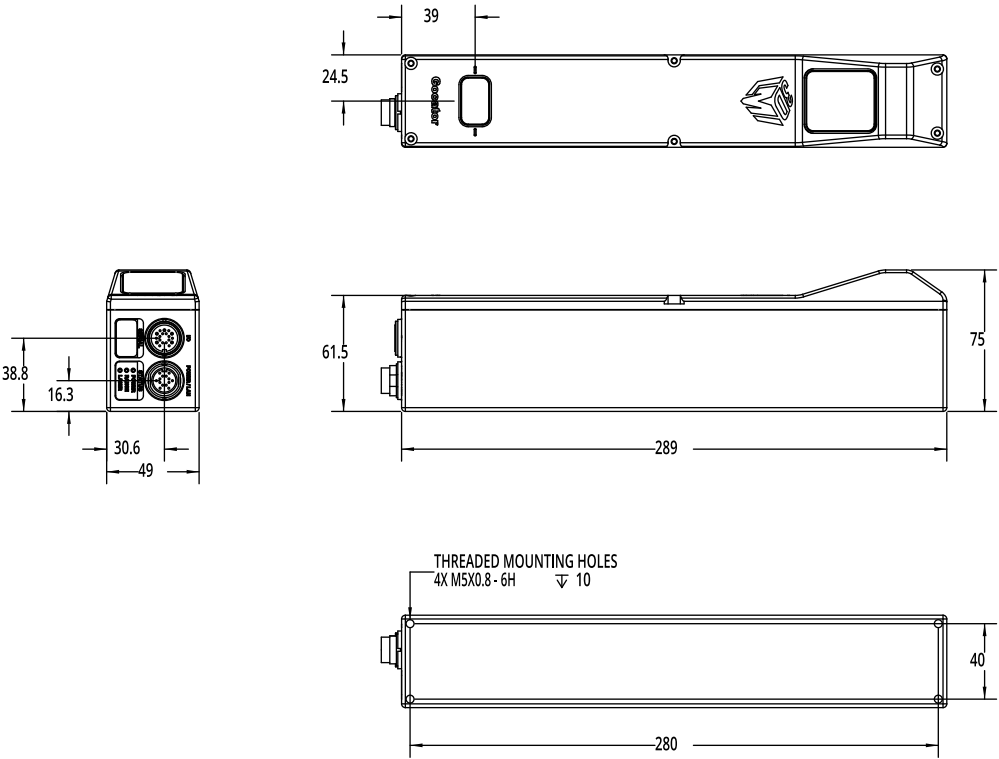


Gocator 1390 (Top Mount Package)

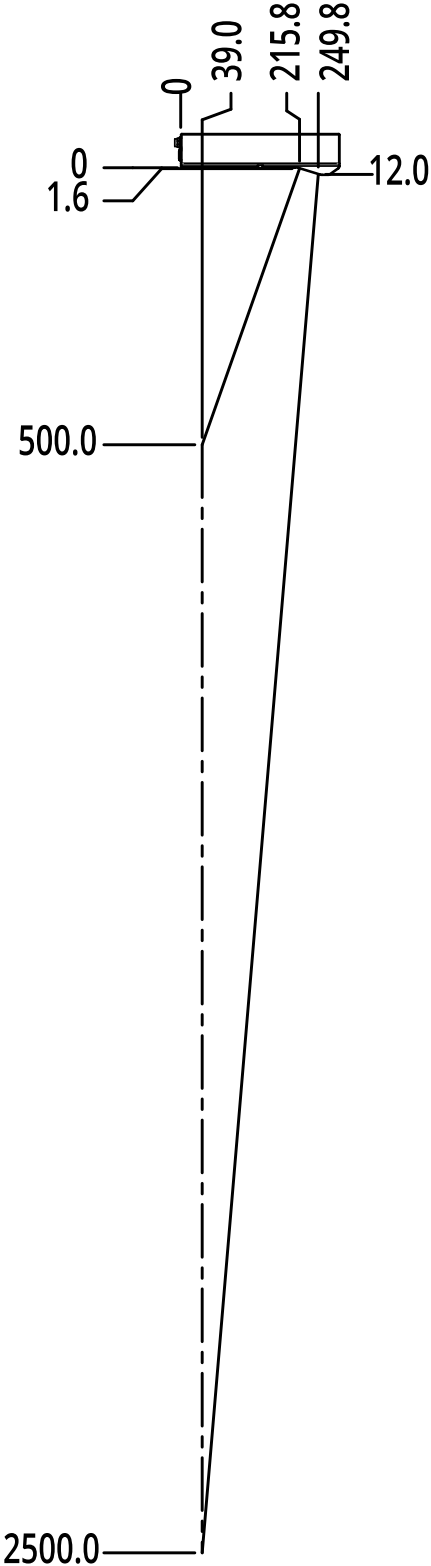
Field of View / Measurement Range



Dimensions



Envelope



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